Thinking on Its Feet: Terrain-Aware Hierarchical Control with Proprioceptive Feedback for Humanoid Navigation

Yuhai Wang[1], Chenghao Wang[1], Xianyao Li[2], Zening Teng[1], Xiao Hu[1], Rongxuan Zhou[1], Alireza Ramezani[1], Jing Du[2], and Yang Ye[1,*]

Project Website: https://yuhaiw.github.io/TOIF/

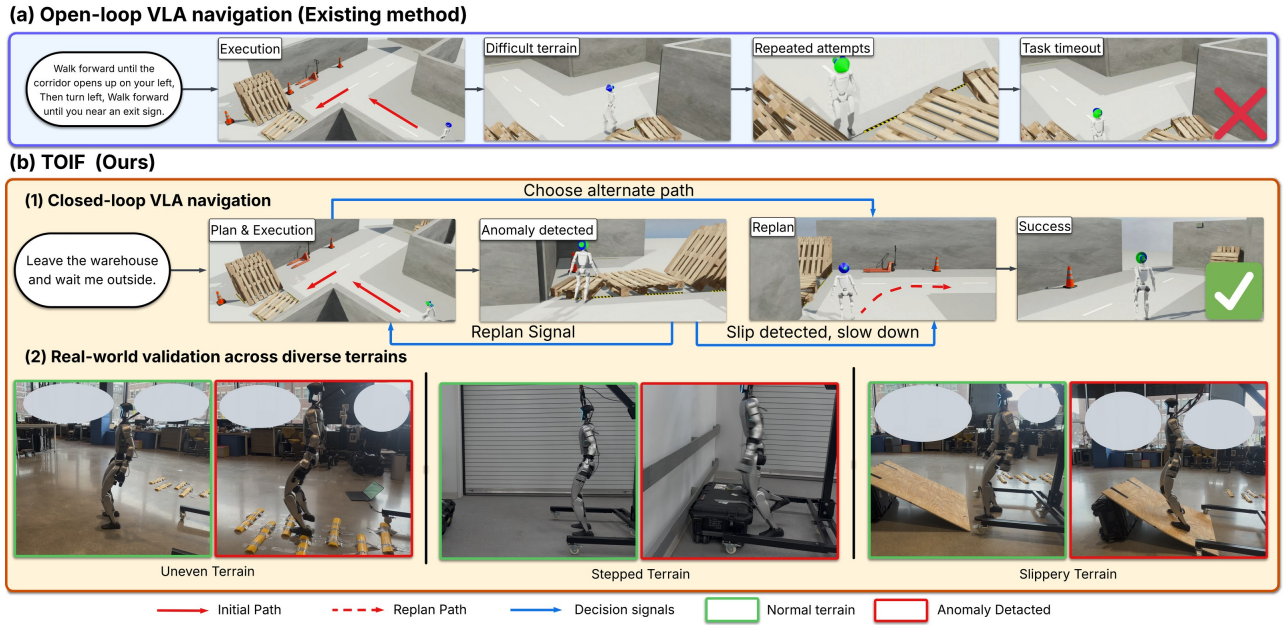


Fig. 1: Motivation of our approach. (a) Open-loop VLA navigation commits to a fixed plan and fails upon encountering challenging terrain. (b) TOIF closes this loop: (1) terrain anomalies are detected via proprioceptive sensing, triggering replanning to an alternative route and modulating velocity to reduce risk; (2) the approach is validated on a physical Unitree G1, achieving reliable anomaly detection on uneven, stepped, and slippery terrains. Green and red borders indicate normal traversal and detected anomalies, respectively.

***Abstract*— Vision-Language-Action (VLA) models have advanced robotic navigation by unifying perception and planning, yet many existing systems lack explicit feedback on locomotion feasibility and cannot adapt high-level plans when execution becomes unsafe. This disconnect can lead to navigation failures on challenging terrains where visually plausible paths prove physically infeasible. We propose Thinking on Its Feet (TOIF), a three-tier hierarchical control framework that closes the loop between low-level proprioceptive sensing and high-level planning for terrain-aware humanoid robot navigation. At the low level, an RL locomotion policy is augmented with a multi-detector anomaly system that monitors terrain-dependent locomotion difficulties using compact proprioceptive features. Detected anomalies are accumulated through leaky integrators and translated into structured natural-language feedback, allowing the high-level planner to reconsider routes based on the robot's physical experience. At the mid level, a VLA model generates velocity commands that bridge high-level waypoints and low-level execution. We further introduce a risk-aware velocity modulation mechanism that reduces speed based on real-time anomaly scores. We evaluate TOIF in Isaac Lab simulation and on a physical Unitree G1 humanoid robot across challenging terrain conditions. TOIF achieves navigation success rates of 65% and 60% in two simulated scenarios, outperforming the evaluated baselines. Hardware experiments further demonstrate the feasibility of proprioceptive anomaly detection and feedback-driven replanning on a physical humanoid robot.**

I. INTRODUCTION

Vision-Language-Action (VLA) models have recently emerged as a promising paradigm for embodied intelligence, enabling robots to interpret natural-language instructions and translate visual observations directly into navigation behaviors [1]–[6]. By jointly reasoning over perception, language, and action, these models provide an intuitive interface for human–robot interaction and demonstrate strong semantic

[1]Northeastern University, Boston, MA, USA.
[2]University of Florida, Gainesville, FL, USA.
*Corresponding author: Yang Ye. Email: y.ye@northeastern.edu

generalization across tasks and environments. Recent systems such as NaVILA [4] convert language instructions into metric navigation commands for legged robots, while approaches such as OmniVLA [5] extend multimodal conditioning to support diverse goal specifications and embodiments. These developments suggest that large-scale vision–language models may serve as a powerful foundation for general-purpose robot navigation.

Despite this progress, many VLA-based navigation systems lack explicit feedback on locomotion feasibility at the planning level [7]. High-level models generate navigation commands based on visual observations and semantic reasoning, but they remain unaware of how well the robot physically executes these commands on the underlying terrain. In practice, a path that appears traversable from visual observations may become infeasible due to slippery surfaces, uneven terrain, or unexpected steps; because locomotion feedback is not integrated into high-level decision making, robots may continue to follow an unsafe or blocked route even after execution failures occur.

Humans naturally rely on feedback from their physical interaction with the environment when navigating challenging terrain. For example, when stepping onto ice, a person immediately senses the loss of traction and adjusts their gait before it becomes visually apparent [8], potentially deciding to reroute entirely. Indeed, several hazard classes, most notably low-friction surfaces, produce little or no exteroceptive signature and reveal themselves only through physical interaction; proprioceptive sensing therefore complements, rather than replaces, visual perception. In contrast, many robotic navigation systems treat locomotion and planning as loosely coupled modules: locomotion controllers react locally to disturbances, while high-level planners remain unaware of execution outcomes [9]. As a result, execution feedback rarely influences global navigation decisions; incorporating such feedback into high-level reasoning can therefore enable safer and more adaptive navigation.

To address this challenge, we present **Thinking on Its Feet (TOIF)**, a hierarchical framework that closes the loop between low-level proprioceptive sensing and high-level semantic planning. TOIF consists of three levels: a high-level planner that reasons over a semantic scene graph using an LLM; a mid-level VLA that converts natural-language instructions and visual observations into velocity commands; and a low-level RL locomotion policy. Importantly, TOIF integrates the three control levels through a bidirectional feedback pathway: proprioceptive anomaly detectors identify terrain-induced failures, a leaky-integrator scheme separates sustained hazards from transient noise, and the detected anomalies are translated into natural-language context that enables the planner to replan around the failure. This design reduces exposure to hazardous terrain: speed is reduced as anomaly confidence rises, limiting the severity of the initial encounter, and detected hazards are excluded from subsequent routes when replanning is triggered.

Our main contributions are:

- A **three-tier hierarchical architecture with bidirectional information flow** that couples downward command propagation with upward proprioceptive feedback via natural-language terrain context.
- A **risk-aware terrain adaptation mechanism** that combines sigmoid-based velocity modulation with leaky-integrator replan triggers, using terrain-specific time constants to balance responsiveness against false-positive suppression.
- A **multi-detector proprioceptive anomaly system** that uses compact, task-specific feature subsets for each terrain type, validated both in simulation and on a real Unitree G1 with different sensor configurations.

II. RELATED WORK

A. VLA Models and Hierarchical Locomotion

VLA models unify perception, language understanding, and action generation within a single architecture. For navigation, VLA models take egocentric RGB frames and natural-language instructions as input and produce velocity commands or waypoints. NaVILA [4] fine-tunes a VILA-based VLM on Room-to-Room instructions blended with spatial QA data, outputting mid-level language commands executed by a separate RL locomotion policy. This two-level design decouples the inference rate of the VLM ($\sim$1 Hz) from the real-time locomotion loop and enables cross-embodiment transfer. OmniVLA [5], trained on over 9,500 hours of navigation data across ten platforms, further extends multi-modal goal conditioning to accept language, 2D poses, and egocentric images.

Compared with classical planners such as A* or RRT*, VLA-based navigation requires no pre-built map and can interpret free-form language goals. Conversely, RL-based locomotion policies [10]–[12] excel at robust low-level control over challenging terrain but lack semantic understanding and long-horizon planning. Hierarchical approaches such as Walk These Ways [13] bridge this gap by conditioning locomotion on velocity commands, providing a motor interface for higher-level planners.

TOIF adopts a VLA as the mid-level navigator and an RL policy as the low-level controller, but critically closes the loop with bidirectional feedback: proprioceptive anomalies propagate upward to trigger semantic replanning, while revised plans propagate downward to redirect locomotion—a mechanism absent from existing VLA navigation systems and hierarchical locomotion frameworks.

B. Safety-Aware Navigation and LLM-Based Replanning

Classical planners such as A* [14] and RRT* [15] guarantee collision-free paths given known maps but assume static, fully observable environments and provide no mechanism for execution-time recovery. Model-based safety monitoring—including contact detection [16] and impedance-based collision response [17]—can trigger reflexive behaviors, yet these reactions are local and do not inform route selection. Learning-based approaches extend feasibility awareness: proprioceptive terrain classifiers modulate traversability costs for quadruped planners [11], [18], and TOP-Nav [19] closes the

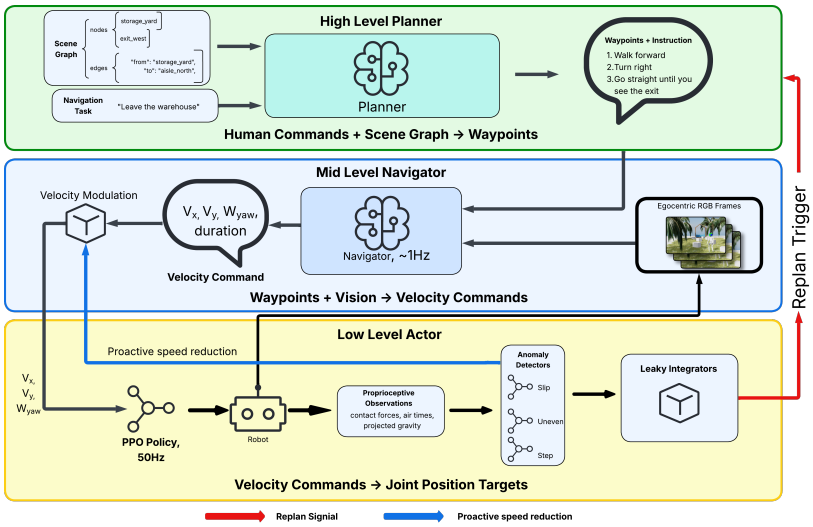


Fig. 2: TOIF system architecture. Commands flow downward through scene-graph planning, VLA navigation, and RL locomotion. Proprioceptive feedback flows upward through anomaly detection, leaky integration, and semantic translation, enabling the planner to replan with terrain-aware context.

loop between vision-based terrain estimation and proprioceptive feedback to correct path costs online. However, these systems operate within a fixed geometric planning framework and cannot replan at the semantic level when an entire route becomes infeasible.

Foundation models offer a complementary path toward adaptive replanning. SayCan [20] grounds LLM plans in learned affordances, Inner Monologue [21] feeds textual success/failure signals back to an LLM for manipulation replanning, and LLM-Planner [22] uses visual observations for few-shot embodied replanning. However, these approaches rely on high-level feedback—binary outcomes, scene descriptions, or human corrections—rather than low-level proprioceptive signals. To the best of our knowledge, no prior work feeds proprioceptive terrain feedback (e.g., slip detection, gait anomalies) into an LLM planner for semantic rerouting. TOIF bridges this gap by translating proprioceptive anomaly signals into natural-language context for LLM-based replanning, enabling terrain-aware rerouting grounded in physical experience rather than visual observation alone.

III. METHODOLOGY

A. System Overview

TOIF operates as a three-tier cascade (Fig. 2). An LLM planner receives a scene graph $\mathscr{G}$ and a navigation task $\mathscr{T}$ and outputs a waypoint sequence with natural-language locomotion instructions (high level, on demand). A VLA model converts the current instruction and an egocentric image sequence into velocity commands $(v_x, v_y, \omega_{\text{yaw}})$ (mid level, $\sim$1 Hz). An RL policy executes locomotion while a multi-detector anomaly system monitors proprioceptive signals (low level, 50 Hz).

The key architectural innovation is **bidirectional information flow**. In the downward direction, the high-level plan is decomposed into mid-level instructions, which are in turn converted into low-level velocity targets. In the upward direction, the anomaly detectors aggregate proprioceptive signals into a structured feedback message $\mathscr{F}$, which modulates mid-level control and prompts the high-level planner to replan.

B. Scene Graph Representation

The environment is encoded as a scene graph $\mathscr{G} = (\mathscr{V}, \mathscr{E})$, where nodes $v \in \mathscr{V}$ represent navigable regions annotated with semantic type and spatial position, and edges $e \in \mathscr{E}$ encode connectivity with directional information. The scene graph contains no prior knowledge of terrain hazards. This ensures that terrain-aware behavior emerges entirely from runtime proprioceptive feedback rather than privileged map information and provides a fair basis for comparison with geometric planners.

C. High-Level: Scene-Graph-Guided Semantic Planning

The high-level planner is an LLM that maps a navigation task $\mathscr{T}$ and scene graph $\mathscr{G}$ into an ordered sequence of waypoints with natural-language locomotion instructions for the mid-level controller. The LLM receives $\mathscr{G}$ as serialized nodes and edges with directional annotations, all simple paths from the current location of the robot to the destination, and the egocentric frame of the robot (the facing direction and the relative left/right/forward/behind mapping). A structured prompt constrains the output to atomic commands (e.g., "Walk forward," "Turn left") paired with a waypoint sequence drawn exclusively from valid node IDs in $\mathscr{G}$.

Feedback-Augmented Replanning. When low-level feedback triggers replanning (Sec. III-E.3), the planner infers which scene-graph edge the robot was traversing when the anomaly occurred and adds this edge to a persistent blocked-edge list that accumulates across replans within a task. The feedback message $\mathscr{F}$ is serialized into the planner prompt as structured natural-language context that reports the anomaly type with its detection confidence, per-detector

risk levels, the triggering reason, and the current pose of the robot, together with the accumulated blocked edges (e.g., "Anomaly type: STEP (confidence 85%); blocked: storage_yard $\leftrightarrow$ exit_west"). While the detector scores are produced synchronously with the 50 Hz control loop and consumed continuously by velocity modulation (Sec. III-F), the upward channel is event-triggered: $\mathscr{F}$ is assembled and sent only when a replan trigger fires, so the LLM is queried on demand rather than at a fixed rate. All remaining simple paths that avoid blocked edges are enumerated and presented to the LLM, which selects an alternative route; this mechanism prevents cyclic rerouting. The LLM serves as the high-level interface for selecting valid routes and generating navigation instructions. The key role of proprioceptive feedback is to update route feasibility during execution, rather than to improve the underlying graph-search algorithm.

D. Mid-Level: Vision-Language-Action Navigation

The mid-level controller translates waypoint-level instructions into metric velocity commands for the low-level RL policy. We adopt NaVILA [4], a VLA model that takes a sequence of egocentric RGB frames and a natural-language instruction as input and outputs a velocity tuple $(v_x, v_y, \omega_{\text{yaw}})$ with an execution duration. Because VLA inference runs at $\sim$1 Hz while locomotion control requires 50 Hz, the velocity command is held for the specified duration, bridging the frequency gap without blocking the control loop. The mid-level controller is stateless with respect to the navigation plan—it receives only the current instruction and visual observations—so the high-level planner and low-level policy can be modified independently.

E. Low-Level: RL Locomotion and Proprioceptive Feedback

The low level executes locomotion commands while monitoring execution quality through proprioceptive anomaly detection.

1) Reinforcement Learning Policy: We employ a PPO-based locomotion policy trained in Isaac Lab for the Unitree G1 humanoid (29 DoF). The policy receives proprioceptive observations—base angular velocity, projected gravity, joint positions and velocities, commanded velocity, and action history—and outputs joint position targets at 50 Hz.

2) Proprioceptive Anomaly Detection: Terrain hazards such as slippery surfaces or unexpected steps are often invisible to a forward-facing camera yet produce clear proprioceptive signatures—abnormal foot contact forces, prolonged aerial phases, or sudden changes in projected gravity. We exploit this observation by training lightweight anomaly detectors on nominal flat-terrain proprioceptive data; at runtime, deviations from the learned nominal distribution indicate terrain anomalies.

Each terrain type k (slippery, uneven, stepped) has a dedicated detector f_k that operates on a compact contact-feature subset selected through ablation (Sec. V-A) and flags anomalies via $\hat{y}_k = \mathbf{1}[s_k(\mathbf{x}) > \tau_k^*]$, where τ_k^* is an F1-optimal threshold. Majority voting over a sliding window of 15 frames (0.3 s) filters transient false positives while preserving sub-second response time.



Fig. 3: **S2: Rocky Beach scenario.** The robot navigates an uneven coastal terrain with randomly placed rock formations. No alternative route is available; the robot must continuously adapt its velocity and gait to traverse the single feasible path to the goal.

3) Replan Triggering: A two-tier mechanism converts detections into upward replan signals. A replan is triggered *immediately* when the body orientation or height violates predefined safety bounds (e.g., $|\theta_{\text{pitch}}|$ or $|\theta_{\text{roll}}| > 0.5$ rad, or $h < 0.4$ m). For ML-based detections, terrain-specific leaky integrators accumulate evidence over time:

$$I_k(t) = \max\big(0,\, I_k(t-1) + \hat{y}_k(t) \cdot \Delta t - \lambda_k \cdot \Delta t\big). \tag{1}$$

A replan is triggered when $I_k(t) > \eta_k$ for any k, followed by a 2.0 s cooldown that prevents cascading triggers. Each terrain type uses a distinct (η_k, λ_k) pair tuned to its failure dynamics—e.g., step failures warrant the most sensitive setting due to the high risk of falling.

F. Risk-Aware Velocity Modulation

Rather than waiting for anomalies to trigger replanning, TOIF continuously adjusts speed through velocity modulation as a first line of defense. For each detector k with score s_k and threshold τ_k^*, the risk estimate is $r_k = \sigma(\alpha(s_k - \tau_k^*)/|\tau_k^*|)$ with $\alpha = 8$. The velocity scaling factor is:

$$\gamma = \min_k \Big[\gamma_k^{\min} + (1 - \gamma_k^{\min}) \cdot (1 - r_k)\Big], \tag{2}$$

where $\gamma_k^{\min}$ is a terrain-specific minimum speed fraction, and the modulated command is $\mathbf{v}' = \gamma \cdot \mathbf{v}$ with exponential smoothing ($\beta = 0.3$) to prevent discontinuities. This design creates a two-stage defense: velocity modulation provides a continuous, graded response, while replanning handles cases in which speed reduction alone is insufficient.

IV. EXPERIMENTAL SETUP

A. Simulation Environment

All simulation experiments are conducted in NVIDIA Isaac Lab [23] with the Unitree G1 humanoid robot. The low-level locomotion policy is trained via Proximal Policy Optimization (PPO) [24] on mixed terrain and deployed zero-shot on unseen terrain configurations. We construct two evaluation scenarios of increasing complexity:

- **S1: Warehouse Exit** (Fig. 1). A T-shaped warehouse with two exit doors. After committing to one exit, the robot encounters an unexpected obstacle—a fallen wooden pallet (20 cm in height) blocking the doorway.

TABLE I: Low-level RL locomotion policy configuration for the Unitree G1 (29 DoF). The same observation and action spaces are used in both simulation and real-world deployment.

Parameter	Value
Observation Space	
Base angular velocity	3D
Projected gravity	3D
Velocity commands $(v_x, v_y, \omega_{\text{yaw}})$	3D
Joint positions (relative)	29D
Joint velocities	29D
Last actions	29D
Total	**96D**
Action Space & Control	
Output	29D joint position targets
Control frequency	50 Hz (sim & real)
Training	
Algorithm / Network	PPO / MLP [512, 256, 128]
Parallel environments	4096
Training iterations	36,500
Sim-to-Real Transfer	
Sim2Sim validation	MuJoCo
Deployment interface	Unitree SDK 2

The robot must detect the traversal failure through proprioceptive feedback, trigger replanning, and reroute to the alternative exit.

- **S2: Rocky Beach** (Fig. 3). An 18 m $\times$ 8 m coastal terrain combining uneven ground with scattered rigid rocks of varying size. The robot must reach a goal 18 m away. Unlike pure rough terrain, the rocks act as localized obstacles that can block forward progress, requiring reactive replanning to circumvent blocked paths.

S1 tests *detection-triggered semantic rerouting* in response to an unmodeled structural obstacle, while S2 tests *continuous terrain adaptation with localized replanning* in the absence of pre-defined alternative paths.

B. Real-World Deployment

The low-level locomotion policy is transferred to the physical robot following a sim-to-sim-to-real pipeline built on the Unitree RL Lab framework [25]. The PPO policy trained in Isaac Lab is first validated in MuJoCo to verify simulator-agnostic behavior and then deployed on the onboard computer via the Unitree SDK 2. Table I summarizes the policy configuration; no additional domain adaptation is required.

To validate practical applicability, we deploy the anomaly detection component on a physical Unitree G1 and evaluate its performance on three terrain types constructed in a laboratory environment: (1) **uneven terrain** using multiple speed bumps arranged to create surface irregularity, (2) **stepped terrain** using 40 cm-high toolboxes as obstacles, and (3) **slippery terrain** using wooden boards inclined at 30–45°. In contrast to the simulation setting, the real-world detectors are trained directly on manually annotated physical robot data, with an 80/20 temporal split to prevent data leakage. The high-level planning and mid-level VLA components operate as in simulation, because they depend only on semantic scene graphs and visual observations, which transfer naturally to real environments.

C. Metrics

For the simulation experiments (S1 and S2), each condition is evaluated over $N = 20$ trials per scenario. We report mean $\pm$ standard deviation for all metrics:

- **Success Rate (SR)** $\uparrow$: Percentage of trials in which the robot reaches the goal within the time limit (120 s).
- **Falls** $\downarrow$: Number of falls per trial, defined as $h < 0.3$ m or $|\theta_{\text{pitch/roll}}| > 1.0$ rad sustained for ≥ 1.0 s.
- **Completion Time** $\downarrow$: Time to reach the goal (successful trials only).
- **Replans**: Number of high-level replans triggered per trial (reported only for methods with replanning capability).

D. Baselines and Ablations

1) Baselines (Table II): We compare TOIF against three baselines:

- **RRT* + Heuristic** [15]: Classical geometric planning on an occupancy map derived from the same scene graph used by TOIF. Waypoint tracking uses a proportional yaw controller at constant forward velocity (0.6 m/s). This baseline lacks both predictive trajectory optimization and terrain awareness.
- **RRT* + DWA** [26]: Extends the preceding baseline with Dynamic Window Approach local planning. DWA samples velocity commands within dynamic constraints, performs 1 s forward rollouts, and selects the command minimizing a weighted multi-objective cost. This baseline adds predictive lookahead but still lacks runtime terrain feedback.
- **NaVILA End-to-End** [4]: The mid-level VLA generates velocity commands directly from visual observations and language instructions, without hierarchical planning or proprioceptive feedback. This baseline isolates the contribution of our closed-loop feedback architecture.

All methods operate on the same scene graph without prior terrain-hazard annotations. The advantage of TOIF arises solely from runtime proprioceptive feedback rather than from privileged environmental information.

2) Ablation Study (Table III): To isolate the contribution of each TOIF component, we evaluate three ablated variants on both scenarios:

- **w/o Velocity Modulation**: Anomaly detection triggers replanning, but speed is not proactively reduced before failure.
- **w/o Anomaly Detection**: No ML-based terrain detection; only stability-based reflexes (pitch/roll/height thresholds) can trigger feedback.

TABLE II: Navigation Performance Comparison (20 trials per condition, mean ± std).

Scenario	Method	SR (%) ↑	Time (s) ↓	Falls ↓	Replans
S1: Warehouse Exit	RRT* + Heuristic	40	71.77 ± 12.98	0.5 ± 0.6	—
	RRT* + DWA	45	26.73 ± 8.58	0.0 ± 0.0	—
	NaVILA E2E	40	49.04 ± 7.17	0.2 ± 0.4	—
	TOIF (Ours)	65	37.34 ± 12.10	0.0 ± 0.0	0.80 ± 0.40
S2: Rocky Beach	RRT* + Heuristic	20	42.88 ± 12.76	0.8 ± 0.4	—
	RRT* + DWA	35	35.70 ± 2.08	0.60 ± 0.52	—
	NaVILA E2E	35	68.09 ± 10.20	0.65 ± 0.49	—
	TOIF (Ours)	60	84.81 ± 16.63	0.40 ± 0.52	0.50 ± 0.53

TABLE III: Ablation Study (20 trials per condition). Each row removes one module from the full system.

Scenario	Variant	SR (%) ↑	Time (s) ↓	Falls ↓	Replans
S1: Warehouse Exit	**TOIF (Full)**	65	37.34 ± 12.10	0.0 ± 0.0	0.8 ± 0.4
	w/o Velocity Modulation	50	39.0 ± 6.9	0.0 ± 0.0	0.9 ± 0.3
	w/o Anomaly Detection	20	20.5 ± 1.5	0.3 ± 0.5	—
	w/o High-Level Planning	15	44.2 ± 9.8	0.1 ± 0.3	—
S2: Rocky Beach	**TOIF (Full)**	60	84.81 ± 16.63	0.40 ± 0.52	0.50 ± 0.53
	w/o Velocity Modulation	50	67.81 ± 10.23	0.50 ± 0.50	0.45 ± 0.51
	w/o Anomaly Detection	25	64.62 ± 11.53	0.75 ± 0.44	—
	w/o High-Level Planning	35	63.20 ± 8.85	0.65 ± 0.49	—

- **w/o High-Level Planning**: Anomaly detection and velocity modulation are active, but no semantic replanning occurs—the robot can slow down but cannot reroute.

Each ablation removes exactly one module, isolating its marginal contribution to overall system performance.

V. RESULTS AND ANALYSIS

A. Feature Selection Ablation

We validate task-specific feature selection using simulation slip detection as a representative case. The full 107D proprioceptive feature set achieves only AUC = 0.66, because joint-state features introduce noise arising from the compensation dynamics of the RL policy. A compact 11D contact-feature set (foot forces, air times, projected gravity) achieves AUC = 0.94, confirming that contact signals encode terrain interaction before policy compensation attenuates the signal.

B. Navigation Performance

Table II compares TOIF against three baselines on both scenarios.

TOIF achieves the highest success rate in both scenarios. On S1 (Warehouse Exit), TOIF reaches 65% SR, outperforming RRT*+DWA (45%), NaVILA E2E (40%), and RRT*+Heuristic (40%). On S2 (Rocky Beach), TOIF achieves 60% SR versus 35% for both RRT*+DWA and NaVILA E2E.

Baselines fail for complementary reasons. NaVILA E2E commits to its initial path and cannot reroute upon encountering the fallen pallet on S1. RRT* variants route through hazardous terrain without adaptation, yielding the highest fall counts (0.8 per trial for RRT*+Heuristic on S2). Local planning with DWA improves over heuristic tracking on S1 (45% vs. 40%) but still lacks terrain feedback.

TOIF trades time for safety. Completion times on S2 are higher than those of the baselines (84.8 s vs. 35.7–68.1 s), reflecting velocity modulation and replanning detours. However, this trade-off yields the lowest fall rates and substantially higher task completion.

C. Ablation Study

Table III isolates the contribution of each TOIF component on both scenarios.

Anomaly detection is the most critical component. Removing ML-based terrain detection causes the largest SR drop: $65\% \rightarrow 20\%$ on S1 and $60\% \rightarrow 25\%$ on S2. Without learned detectors, the system relies solely on stability-monitor reflexes (pitch/roll/height thresholds), which trigger only in near-failure states, increasing the number of falls to 0.3 and 0.75, respectively.

Velocity modulation provides consistent gains. Removing proactive speed reduction decreases SR by 10–15 percentage points while increasing falls on S2 ($0.40 \rightarrow 0.50$), confirming that graded speed adaptation complements binary replan decisions on sequential terrain segments.

High-level planning is critical for multi-path scenarios. Without semantic replanning, SR drops sharply on S1 ($65\% \rightarrow 15\%$): the robot detects the obstacle but remains stuck on the blocked path. On S2, the effect is more moderate ($60\% \rightarrow 35\%$), as the open layout offers fewer routing alternatives.

D. Real-World Validation

Table IV summarizes real-world detection performance. The stepped-terrain detector achieves the highest AUC (0.942, XGBoost [27]) using a single IMU feature, because step-over failures produce a characteristic yaw disturbance. The uneven-terrain detector reaches AUC = 0.874 (AdaBoost, Fig. 5), while slippery terrain remains the most challenging

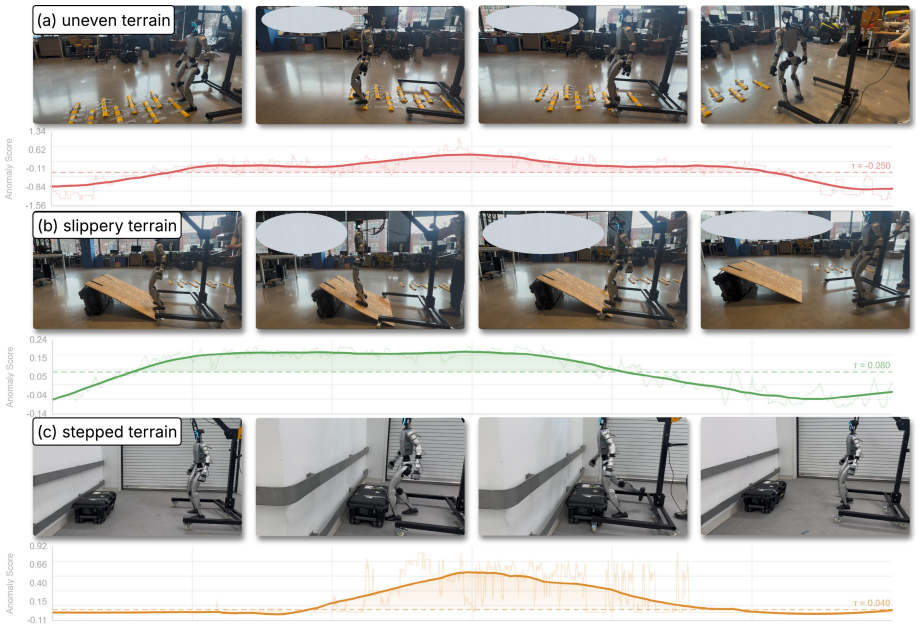


Fig. 4: Real-world anomaly detection on a physical Unitree G1 across three terrain types. Each row shows robot snapshots (top) and temporal anomaly scores (bottom). **a:** uneven terrain (τ=−0.250). **b:** slippery terrain (τ=0.080). **c:** stepped terrain (τ=0.040). Bold curves: smoothed scores; light traces: raw output; dashed lines: F1-optimal thresholds.

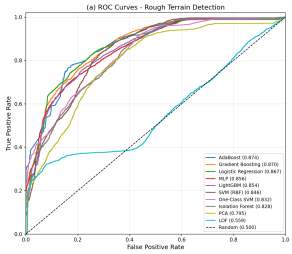


Fig. 5: ROC curves of candidate models for real-world uneven-terrain anomaly detection, trained on physical Unitree G1 data.

to detect (AUC = 0.813, LOF) because gradual gait changes overlap with normal speed variation. Fig. 4 confirms that all three detectors exhibit prompt activation upon terrain entry and reliable decay upon exit—the temporal properties essential for the TOIF feedback loop.

A notable finding is that supervised classifiers (AdaBoost, XGBoost) outperform the one-class methods used in simulation. Real-world nominal data exhibits higher variance, which compresses isolation-based score distributions and favors methods that explicitly model the normal–anomalous boundary. The framework accommodates this naturally by treating model selection as a deployment-specific decision.

Finally, we deploy the complete three-tier pipeline on the physical G1 in a laboratory navigation task: the robot is instructed to reach an exit sign, while a wooden pallet on a hand truck blocks the direct route. Upon encountering the obstacle, the proprioceptive anomaly detectors trigger replanning, and the planner reroutes the robot along an alternative path to the exit (Fig. 6).The robot reached the exit in 3 out of 5 trials, demonstrating the feasibility of the full pipeline on physical hardware.

TABLE IV: Real-World Anomaly Detectors (Unitree G1)

Terrain	Model	Dim	AUC	Key Features
Uneven	AdaBoost	4D	0.874	torso sway, hip torque
Stepped	XGBoost	1D	0.942	IMU yaw orientation
Slippery	LOF	4D	0.813	ankle pos, trunk incl.

E. Discussion

Verification of core hypothesis. The ablation study directly supports our central claim that closing the loop between proprioceptive feedback and semantic planning improves robustness: removing any single module degrades performance, and the complementary failure modes of the baselines—NaVILA E2E lacking replanning, RRT* lacking terrain awareness—confirm that neither perception-only nor planning-only approaches suffice. While classical graph-search methods could also perform the rerouting step, TOIF focuses on incorporating proprioceptive feedback into high-level planning rather than improving the graph-search algorithm itself.

When does each component matter? The results reveal a clear task dependence. High-level replanning is decisive when alternative paths exist (S1) but contributes less in open layouts (S2). Velocity modulation provides its largest benefit on S2, where continuous adaptation matters more than discrete rerouting. Anomaly detection is universally critical as the perceptual foundation for both mechanisms.

Scalability of the detector suite. The detector suite is a modular component of TOIF rather than its final form: the planner, the VLA, and the locomotion policy are agnostic to how anomaly scores are produced, so the per-terrain detectors can be replaced by more general alternatives with-

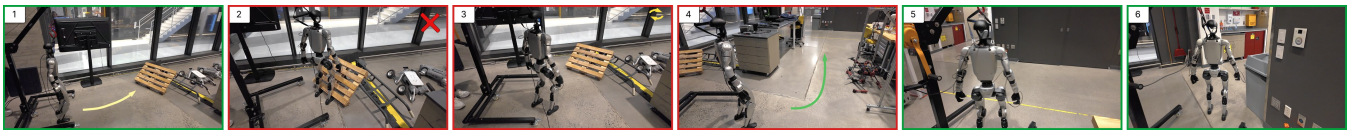


Fig. 6: Full-pipeline deployment on the physical Unitree G1. The robot walks toward the exit sign (1); a wooden pallet blocks the direct route and an anomaly is flagged (2–3); replanning turns the robot toward the alternative route (4), which it follows to the exit (5–6). Green/red borders: normal traversal / detection–replanning phase; yellow/green arrows: initial/replanned route.

out modifying the other tiers. Our experiments validate the closed-loop framework with lightweight classical detectors, which in simulation require only nominal flat-terrain data and F1-calibrated thresholds; unifying detection across terrain types is a natural direction for future work.

Limitations and future work. The hardware evaluation is limited to five trials, and the anomaly detectors require retraining on real-world data rather than transferring directly from simulation. Failures mainly arise from VLA turning errors and discontinuous velocity commands. Future work will explore more generalizable detectors, larger-scale hardware evaluation, and replanning latency analysis.

VI. CONCLUSION

We presented TOIF, a hierarchical framework that closes the loop between proprioceptive sensing and semantic replanning for terrain-aware humanoid navigation. By translating proprioceptive anomaly signals into natural-language feedback, TOIF enables an LLM planner to reroute based on the physical experience of the robot rather than on visual observation alone. Experiments in Isaac Lab and on a physical Unitree G1 show that the closed-loop architecture consistently outperforms open-loop baselines, with ablations confirming the complementary contributions of anomaly detection, velocity modulation, and semantic replanning. Future work will focus on tighter VLA–proprioceptive integration, temporal anomaly features for gradual terrain transitions, multi-modal sensor fusion (e.g., vision and tactile signals) to enrich the scene graph, and a larger-scale evaluation of the complete closed-loop system on physical hardware.

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